

PROJECT IV

GRAVITY MODELS OF TRADE FOR EU PANEL DATA

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Gravity Model · EU Trade Flows · Panel Econometrics

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Introduction

Hsiao (2014) identifies panel data in econometrics as being the most exiting field in econometrics today. In fact, there are three types of data: cross-sectional data, times series data and panel data. The cross-sectional data deals with different units of a data set within a particular time, these could be data on households, individuals and what have you. The time series data further observes the data at a macro level over a period of time unlike the cross-sectional within a particular time (like a year). Panel data, also known as the longitudinal data, envelops the two, cross-sectional and time series data – called the dimensions of the panel data (Pillai, 2007). According to Hsiao (2007), panel data refers to data sets that contains time series observations of variables or individuals. The Panel data is a pool of cross-sectional data sets over multiple time periods. In addition, panel data have two types which are the micro panels and the macro panels. Since this observes the same cross-sectional data variables over a long period of time, the data set has variables which are generalized, common and possess individual behaviors.

Panel data helps address the heterogeneity problems in the data sets and the data analysis, which are the differences across the data sets. Furthermore, following the period of the data set, panel data helps to analyze particular variables over time to ascertain their performance and behavior. It is observed that this kind of data is more informing and very helpful in economic analysis. In regressing panel data set, analysts are able to reduce the collinearity problem in the variables as well as have more degree of freedom. These together ensure effective analyses of data for policy recommendation or good and sound analysis of the data set. The increase in panel data analysis usage is due to its ability to model realistic behavioral hypothesis, as well as challenging methodological problems. More to this, is its ability to capture the complexity of human

behavior, than the other two data types, according to Hsiao (2007). These affirm why it is commonly used in economic literature.

Nonetheless, there are challenges in the use of the panel data for analysis. First is the issue of data collection which is very cumbersome owing to the need to collect data across the data units for a long period of time. Meanwhile, if the data is successfully collected, distortions of the measurement errors are usually observed, as well as some selectivity problems. As a result of this, in some instances, individual in the panel data could be lost. Also, Hsiao (2007) identifies that the need to control the effect of unobserved heterogeneity is a challenge in the panel data methodology.

Other disadvantages are the short time series dimension and its cross-sectional dependence. That notwithstanding, the application of time series data is widely acknowledge following its benefits to data analysis and the outcomes.

In view of these, this paper employs panel data analysis to estimate “gravity models” of trade for EU panel data. In detail, various panel data models will be estimated to analyze the data set. This model estimated in panel data analysis allows for both time variant and time invariant effects, a key observation of the approach. Most of the gravity models of trade either used the cross-section model or the pooled cross section model (Cheng and Wall, 1999). According to Cheng and Wall (1999), these two impose restrictions that the slopes are same across pairs of countries and the same parameter vector for all, respectively, while using the OLS estimator. To address these restrictions, the fixed effects model is applied while handling the country pair heterogeneity (Matyas, 1997).

The work by Serlenga and Shin (2007) also adopts the gravity model of trade for European countries from 1960 to 2001. The panel-based approach was used in the paper due to its ability to model heterogeneity even for country pair effects, unlike the cross-sectional model. The paper acknowledged that the fixed effects estimator was

desired following the assumption that the individual effects were uncorrelated, which was rejected. However, the fixed effects approach is unable to estimate the influences of time-invariant variables like distance. Therefore, for gravity model using the panel data analysis is key since it is important to consistently estimate the time-invariant effects in the variables (Serlenga and Shin, 2007). In order to address the limitation of the OLS approach by Cheng and Wall (2005), they used Hausman-Taylor's instrumental variable estimation technique.

To further add to literature and estimations of the gravity model of trade, this paper employs different estimation techniques to ascertain the outcome of the gravity model under different estimation approaches. Specifically, the EU panel data is used in the paper, after the paper by Serlenga and Shin (2007). The paper begins with discussing the theory behind the gravity model of trade, describes the data and moves on to the estimation techniques under the empirical section. For each of them, the results are interpreted and finally the conclusion which presents the summary and outcomes of the paper.

Theoretical Framework

Gravity model has been remarkably used to describe the determinants of various types of flows, like trade, recreational traffic, migration, and several others (Serlenga and Shin, 2007). In international trade, the model explains the size of trade flows between countries can be determined by conditions for supply from the origin of trade and the demand conditions at the destination. Anderson (1979) asserts that the gravity model is the most successful trade device that has been used to wide range of goods and factors in cross border trade. He adds that it has shown fit even under different conditions of trade. However, the usage of the model was disturbed by the unavailability and unidentified properties.

The gravity model was developed by Tinbergen and Poyhonen, independently (Cheng and Wall, 1999). Cheng and Wall, (1999) opined that the gravity model of trade explains that the value of trade between two countries by assumptions, increases in size when measured by their national income and decreases in cost of transportation between them, as the distance between the trade centers is considered. Anderson (1979) presented in his seminal work that the gravity model is a run on cross-sectional data and sometimes on pooled data as well. In addition, the model stems from pure expenditure system model by constraining it through specifying the share of the accounted national expenditure on tradable, especially for a Cobb Douglas Expenditure System.

The model is observed to be derived from different structural models like “Ricardian models, Heckscher–Olin (HO) models and Increasing Returns to Scale (IRS) models of the New Trade Theory”, according to Serlenga and Shin (2007). However, the model cannot test how valid these trade models are against each other. Yet it has been successful in its estimations because of the ability to adopt most of the empirical phenomena that are identified in international trade. More so, since the gravity model is able to address empirical issues a lot by reconciling it with theory, it is referred to as the “workhorse of empirical studies of regional integration” by Eichengreen and Irwin (1997).

Finally, owing to the challenge of the cross-sectional OLS in estimating gravity models, the panel data analysis is used. The former could not estimate the heterogeneity issues especially for bilateral trade for the gravity model. Therefore, panel data approach which caters for the heterogeneity problems is identified as best for the gravity model.

Data Description

The data for this project was obtained from the Journal of Applied Econometrics Data Archive. It is important to note that this data is the same as that used by Serlenga and Shin (2007), who published their findings in the Journal of Applied Econometrics. The data was obtained from the Journal's data archive, volume 22, issue number 2, 2007. The data is sampled across ninety-one different countries over a 42-year period (1960 to 2001), with six time-varying regressors, three time-invariant regressors, and six time-specific common factors. The data sampled across ninety-one different countries represents the cross-section portion of the panel data, while the data over the forty-two-year period represents the time series portion of the panel data (see table 2). The ninety - one observations include country pairs in the European Union that trade with one another, as a result of the gravity model's double index version. The country-pairs observations include all the European Union countries. The sum of logged exports and imports bilateral trade flow is the dependent variable (TRADE). The country samples include all 15 EU countries: Austria, Belgium, Denmark, Finland, France, Germany, Greece, Ireland, Italy, Luxemburg, Netherlands, Portugal, Spain, Sweden, and the United Kingdom, with Belgium and Luxemburg treated the same, yielding 91 pairs.

Furthermore, the dataset included variables like RERT, FTRADE, FGDP, FSIM, FRLF, and FRER. RERT is the logarithm of the real exchange rate between European currencies and the US dollar, whereas FTRADE, FGDP, FSIM, FRLF, and FRER are the time-specific common factors (individual means) of the variables TRADE, GDP, SIM, RLF, and RER, respectively.

The six time-varying regressors include the sum of the logged real GDP, the similarity between two trading countries (SIM), relative factor endowments (RLF), the logged bilateral real exchange rate (RER), a dummy equal to 1 when both countries are

members of the European Community (CEE), and a dummy equal to 1 when both countries adopt the common currency. The three time-invariant regressors are the geographical distance between capital cities (DIST), a dummy equal to 1 when the trading partners share a border (BOR), and a dummy equal to 1 when both countries and trading partners speak the same language (LAN). Other variables include logged real exchange rates between European currencies and the US dollar (RERT) and time-specific common factors of five variables (TRADE, GDP, SIM, RLF, and RER). The first forty -two rows represent all-time observations for the first country pair, the next forty-two represent all-time observations for the second country pair, and so on until the 91st country pair of the entire sample. The xtsum command was used to show the overall data set, between, and within the variability of the panel data to confirm the time-invariant variables(Table 2). Table 1 displays the descriptive statistics for the outcome.

There were no differences between the variables RERT FTRADE, FGDP, FSIM, FRLF, and FRER. As there are no differences between the individual units, these variables are dropped or excluded from this study. The study employs time-invariant variables as well as the remaining variables that vary between and within the panel.

Table 1

```
. describe $id $tt $ylist $xlist
```

variable name	storage type	display format	value label	variable label
id	byte	%10.0g		CROSSID
tt	long	%ty	tt	OBS
TRADE	double	%10.0g		TRADE
SIM	double	%10.0g		SIM
RLF	double	%10.0g		RLF
RERT	double	%10.0g		RERT
RER	double	%10.0g		RER
CEE	byte	%10.0g		CEE
GDP	double	%10.0g		GDP
EMU	byte	%10.0g		EMU

```
. summarize $id $t $ylist $xlist
```

Variable	Obs	Mean	Std. Dev.	Min	Max
id	3,822	46	26.27129	1	91
TRADE	3,822	4.894827	1.899349	-1.036677	9.202658
SIM	3,822	-1.33562	.6574685	-3.586714	-.6931475
RLF	3,822	8.401505	1.23432	1.27593	10.20999
RERT	3,822	4.179426	.5647658	3.2835	4.8836
RER	3,822	-.9609118	2.852824	-8.865199	5.623883
CEE	3,822	.4358974	.4959387	0	1
GDP	3,822	13.4148	1.015532	10.48128	15.58845
EMU	3,822	.0379383	.191072	0	1

Table 2

[illegible]

Empirical Section

MODEL SPECIFICATION

In general, the specification of a regression model is based on theoretical rather than empirical considerations. In this case, we can estimate the model using cross-sectional Ordinary Least Squares (OLS). However, one disadvantage of using the cross-sectional OLS is that it does not account for any of the heterogeneous characteristics of the bilateral trade relationships between all of these different pairs of countries. Because it has been argued that cross-section OLS estimation ignores heterogeneous factors, resulting in heterogeneity bias, I use a panel-based approach to estimate different panel models to determine which model is a good fit for the data. The assumption is primarily based on whether the unobserved heterogeneity is correlated or uncorrelated with the variables on the right side. The model specifications to be investigated in the study are presented in the following subsections.

The Pooled Regression

The pooled OLS regression model is one of the models used in panel dataset analysis. It has only one constant term in its coefficients. When all of these assumptions are true, the OLS estimator becomes consistent and efficient. If these assumptions are violated, the pooled OLS estimator becomes inconsistent and unreliable for statistical inference. However, despite switching from a cross-sectional to a panel-based approach, we may be unable to eliminate heterogeneity by using a pooled regression model because it does not account for heterogeneity bias, which may lead to model misspecification.

The pooled model is defined as:

$$Y_{it} = \alpha + X'_{it} \beta + \epsilon_{it} \quad (1)$$

β , and α are time-invariant coefficients.,

where Y_{it} is the dependent variable,

X'_{it} is the set of regressors and are time-invariant coefficients,

ϵ_{it} is the disturbance or error term

i and t are the unit of observation and time period, respectively.

Under classical assumptions, pooled regression is a good estimator. Pooled regression, on the other hand, may not be an efficient estimator because it ignores possible heterogeneity between units by pooling all pairs of countries together. The Pooled Regression Model has a major shortcoming in that it does not distinguish between the various pairs of countries that are our variables of interest. That is, by pooling all the pairs of countries, we deny any heterogeneity or individuality that may exist among them all-time periods. As a result, when we run the pooled regression model, we combine all the pairs of countries while ignoring the possibility of individual heterogeneity among them. If it turns out that the heterogeneity that it ignores varies across countries, then this unobserved heterogeneity causes autocorrelation, and thus any potential effect that is left out of the model will carry across. It is possible to argue that heterogeneity varies across individual units, and that failing to account for this heterogeneity may result in autocorrelation among the individual units.

In this study, I will be concerned with the pooled regression results only if the Hausman test shows that the Random effect model is the best fit to the data. When this actually occurs, additional tests, such as the Bresusch-Pagan Lagrange Multiplier test, will be run to determine whether the Pooled OLS model is more appropriate than the Random Effects model.

Results and Interpretation

I implemented the pooled regression model for this project by drawing all observations together and then estimating the resulting regression model. Stata output for the pooled regression model is shown in (Appendix 2). The Pooled model's R-squared value (0.9067) indicates that the regressors explained approximately 90% of the variation in bilateral trade flows among EU countries. The overall model is significant at 5% level of significant. The p-values of the individual variables are less than 5% significant level we can conclude that each independent variable is significant is explaining the dependent variable (Trade flow). To interpret the coefficients, the SIM(similarity between two trading countries) coefficient implies that a 1% increase in the index that measures the similarity between two countries will increase the estimated bilateral trade flows between the countries by 85% holding other variables constant. The coefficient of the RLF variable indicates that when the relative difference in factor endowment increases by 1%, bilateral trade flows between two countries will increase by about 3%, holding other regressors constant. The bilateral trade volume decreases by about 68 percent as the geographical distance between capital cities increases by 1%, when other regressors are held constant. The remaining variable coefficients can be interpreted in the same manner. As a result, a percentage increase (decrease) in each variable increases (decreases) trade volumes by the variable coefficient in percentage terms.

The pooled results appear to be efficient and consistent, but they are likely biased because the model fails to account for heterogeneity among individual units. The fixed model's heterogeneity test revealed the presence of heterogeneity. The Breusch-Pagan Lagrange multiplier test for random effects was used to choose between the pooled regression and the random effect models. The test is carried out under the null hypothesis of no random effect and the alternative hypothesis of random effects. The outcome (see table 3) suggests that we reject the null hypothesis and accept the

alternative hypothesis of random effect. In that case, we are confident that the pooled regression model is not appropriate for this study.

Table 3 – Breusch Pagen Lagrange Multiplier

Breusch and Pagan Lagrangian multiplier test for random effects		
TRADE[id,t] = Xb + u[id] + e[id,t]		
Estimated results:		
	Var	sd = sqrt(Var)
TRADE	3.607528	1.899349
e	.0853228	.2921007
u	.2660945	.5158435
Test: Var(u) = 0		
	$\chi^2_{(01)} = 41296.29$	
	Prob > $\chi^2 = 0.0000$	

Random Effect Model

The random effect model is basically used when unobserved heterogeneity characteristics are assumed to be uncorrelated with the right-hand side variables. In that case, neither the pooled nor the fixed effect models are suitable. Thus, for the random-effects model, the unobserved individual heterogeneity is assumed uncorrelated with the individual variables. If we assume that the unobserved individual heterogeneity is uncorrelated with the included variables, we can employ the random effects model,

which essentially states that the independent variables have a common mean value for the intercept. That is, because the individual specific effects are distributed independently of the regressors, they should be included in the error term.

The random effect model is defined as:

$$Y_{it} = \alpha + X'_{it} \beta + \mu_{it} + \epsilon_{it} \quad (2)$$

In the model above, μ_{it} is treated as group-specific random effect. The independent variables in the random effects model is specified in such a way that they have common mean value for the intercept. Thus, the individual specific effects are distributed independently of the independent variables and hence they should be included in the error term. The intercept in the model above is treated as random. As previously stated, the choice between the random effect model and the pooled model is determined by the Bresusch-Pagan Lagrange multiplier test. The Hausman test can be used to determine whether there is a statistical difference between fixed effect model and random effect model. If the Hausman test indicates that the random effect model is appropriate and the Bresusch-Pagan Lagrange multiplier test rejects the null hypothesis of zero variance in the group, we can be confident that the random effect is the better estimator of our analysis.

The random effects model uses a simple but very strong assumption to restrict this relationship: namely, that the intercept α_i is a random variable which is not related to X_{it} and ϵ_{it} , in the sense that:

$$E(\alpha_i) = \alpha, \text{Var}(\alpha_i) = \sigma^2_{\alpha} \text{ and } \text{Cov}(\alpha_i, \epsilon_{it}) = 0,$$

all assumed independent of x_{it} . (These moments should be interpreted as conditional on X if the regressors are viewed as random.) Relabeling the variance of ϵ_{it} as $\text{Var}(\epsilon_{it}) \equiv \sigma^2_{\epsilon}$, the original panel data model can be rewritten as $Y_{it} = \alpha + X'_{it} \beta + \mu_{it}$

Where,

$$U_{it} \equiv (\alpha_i - \alpha) + \epsilon_{it}$$

Results and Interpretation

I implemented the random effect regression model and the results from the model explained about 90% of each variation (within, between, and overall) in the dependent variable (see appendix 3).

We can conclude from the significant probability(pvalue) of the individual variables that each independent variable is significant in explaining the dependent variable, trade flow, at the 5% significance level. Furthermore, we see that the F-prob value of 0.0000 is less than a significant value of 0.05. This means that the model as a whole is statistically significant in predicting the expected relationship between bilateral trade flows between countries. To interpret the coefficients, the SIM(similarity between two trading countries) coefficient implies that a 1% increase in the index that measures the similarity between two countries will increase the estimated bilateral trade flows between the countries by 119% holding other variables constant. The coefficient of the RLF variable indicates that when the relative difference in factor endowment increases by 1%, bilateral trade flows between two countries will increase by about 3%, holding other

regressors constant (this is similar to the pooled regression & fixed effect model). The bilateral trade volume decreases by about 53 percent as the geographical distance between capital cities increases by 1%, when other regressors are held constant. The remaining variable coefficients can be interpreted in the same manner. As a result, a percentage increase (decrease) in each variable increases (decreases) trade volumes by the variable coefficient in percentage terms. The Hausman test would be used to determine which model is more appropriate.(Fixed effect or random effect model).

Fixed Effect Model

The fixed effect model is basically used when the unobserved heterogeneity is correlated with the explanatory variables. The fixed-effect model allows for heterogeneity or individuality among different pairs of countries by having its own intercept values. The term "fixed effect" is used here because, while the intercept varies between pairs, it does not vary over time, making it time-invariant. The fixed effect model accounts for heterogeneity across country pairs because the individual or group-specific effect is not observable but is correlated with the right-hand side variables. When this occurs, using the ordinary least square estimator will produce inconsistent and biased results.

The random effect model is defined as:

$$Y_{it} = \alpha_i + X'_{it} \beta + \epsilon_{it} \quad (3)$$

where Y_{it} is the dependent variable,

α_i is the group-specific constant,

X'_{it} is the set of regressors and are time-invariant coefficients,

ϵ_{it} is the disturbance or error term,

i and t are the unit of observation and time period, respectively.

The group-specific constant (α_i) varies across pairs, but it is time-invariant.

The use of a fixed effects model to analyse the relationship between bilateral trade flows between countries is argued to be appropriate. This is due to the fact that different countries have distinct features, and their variable are on the nominal scale of measurement and hence it would be very difficult to observe and quantify. The fixed-effect model's inability to estimate coefficients of time-invariant variables, such as country dummy variables, is a significant limitation. However, in order to better account for heterogeneity effects, these variables must be included or estimated. The fixed effect model is primarily used to control for time-invariant heterogeneity that varies across units.

A drawback of the fixed-effect model is its failure to identify any components of β corresponding to regressors that are constant over time for a given individual; for such coefficients to be identified, stronger conditions on the relation of the individual-specific intercept α_i to the regressors X'_{it} must be imposed.

Results and Interpretation

I implemented the fixed effect regression model for this project and results indicates that the within (time-series dimension), between (cross-section dimension), and overall variations in bilateral trade flows explained by the regressor are approximately 90%,

80%, and 82% respectively(see appendix 4). This result indicates that the fixed effect model accounted for or explained more variations within units. The rho value of the fixed effect estimator indicates that differences between panels account for approximately 87 percent of the variance. The overall model is significant at 5% level of significance(thus F-prob of $0.000 < 0.0500$ alpha level). This means that the entire model is statistically significant in predicting the expected relationship between bilateral trade flows between countries. All the regressors in the model are statistically significant at 5% significance level. The model also rejects the null that fixed effects are jointly zero, indicating that heterogeneity is present. To interpret the coefficients, the SIM(similarity between two trading countries) coefficient implies that a 1% increase in the index that measures the similarity between two countries will increase the estimated bilateral trade flows between the countries by 123% holding other variables constant. The coefficient of the RLF variable indicates that when the relative difference in factor endowment increases by 1%, bilateral trade flows between two countries will increase by about 3%, holding other regressors constant (this is similar to the pooled regression). The remaining variable coefficients can be interpreted in the same manner. As a result, a percentage increase (decrease) in each variable increases (decreases) trade volumes by the variable coefficient in percentage terms. The Hausman test would be used to determine which model is more appropriate.(Fixed effect or random effect model).

First Difference Model

The first difference method estimates panel data by determining the first difference between the dependent and explanatory variables. When the data is first differenced, it

is assumed that this method eliminates unobserved heterogeneity. The first difference method switches data from one period to the next while removing all time-invariant variables. This method has the advantage of removing latent heterogeneity caused by fixed effects or random effects. On the negative side, the model only considers time-varying cross-section units. The first difference estimator is unbiased and consistent under the assumption of strict endogeneity. When the panel data contains only two time periods, the fixed effect and first difference estimators produce identical estimates. Despite the fact that our data was collected over more than two time periods, the first difference method was used in this study solely to compare the results with the other panel data models discussed above. The model is described as follows:

$$\Delta Y_{it} = \Delta C_i + (\Delta X_{it})'\beta + \Delta \epsilon_{it} \quad (3)$$

Where ΔY_{it} first difference in the dependent variable,

ΔC_i first difference for Unobserved heterogeneity,

ΔX_{it} first difference in set of regressors and are time-invariant coefficients

$\Delta \epsilon_{it}$ first difference in the disturbance or error term.

After removing the Unobserved heterogeneity the model will be:

$$\Delta Y_{it} = (\Delta X_{it})'\beta + \Delta \epsilon_{it} \quad (4)$$

Results and Interpretation

I implemented the first difference and the stata output can be found in appendix 5. In the first difference estimator, differences in countries' relative factor endowment (RLF) and trading partners' adoption of common currency (EMU) were found to be insignificant because their pvalues $\{0.938(\text{RLF}), (0.098)\}$ are greater than 5% level of significance. The magnitude of the coefficients is also significantly different from other

model estimates. The findings show that countries' similarity in size (SIM), real exchange rate (RER), trading partners from the same European community (CEE), and real GDPs have a positive effect on total bilateral trade volumes. Finally, the study concludes that the fixed-effect estimator is more appropriate for the data. Despite the fact that both estimators removed time-invariant variables such as BOR, LAN, and DIST, the first difference estimator had a very low R-square value (28%) compared to 81 percent for the fixed-effect estimator.

The Hausman Test

The Hausman test was used to choose between fixed and random effect models. The test aids in determining whether the fixed-effects or random-effects model is more appropriate.

Hypothesis of Hausman Test:

H_0 : The random-effects model is more appropriate.

H_1 : The fixed-effects model is more appropriate.

Results and Interpretation

The Hausman tests also determine whether or not the unique errors are correlated with the explanatory variables. The fixed effect model is consistent under both the null and

alternative hypotheses, whereas the random effect model is inconsistent under the null but efficient under the alternative. The outcome suggests that the null should be rejected. This test result concludes that the fixed-effect model is the best fit for the study's panel data(see appendix 6). We can see from the results that the p-value, 0.001, is less than 5% level of significant and thus we reject the null hypothesis. As a result of the Hausman test, we conclude that the fixed-effects model is more appropriate for explaining variations in bilateral trade.

Because the result indicates that the fixed effect is the correct model, we can now assume that the time-invariant characteristics are unique to each individual unit, and thus the errors should not be correlated. As a result, the effects of time-invariant factors like BOR, LAN, and DIS must be controlled or eliminated.

Diagnostic Test

Following the Hausman test results, and because the fixed effect model was preferred, I conducted diagnostic tests for the fixed model. Using a heteroscedasticity test for the fixed-effect model, I first determined whether the variance of the errors (residuals) is homoscedastic or heteroscedastic.

Hypothesis of the Test:

H_0 : The residuals are homoscedastic

H_1 : The residuals are heteroscedastic

Results and Interpretation

The Stata result for the test is presented in appendix 7. The results of the test indicate that the P-value(0.000) is less than 5% level of significance. Hence the null of homoscedastic would be rejected in favour of the alternative, indicating unequal error variances, particularly in cross-sectional units. Because of the heteroscedasticity, the fixed effect estimator is not the most efficient in this case. As a result, estimating the fixed effect model under the assumption of homoscedasticity may yield inconsistent results.

To account for heteroscedasticity, the fixed model was re-estimated with robust standard errors. The stata results is presented in appendix 8, the results are not significantly different from when the fixed model was estimated with homoscedasticity, except that the robust standard errors and significance level now differ. For example, REF is now statistically significant at 5%.

Testing for cross-sectional dependence/contemporaneous correlation:

Using Pasaran Cross – sectional dependence test.

According to Baltagi, cross-sectional dependence is a problem in macro panels with long time series (over 20-30 years). This is not much of a problem in micro panels (few years and large number of cases). Pasaran CD (cross-sectional dependence) test is used to test whether the residuals are correlated across variables. Cross-sectional dependence can lead to bias in tests results (also called contemporaneous correlation).

H_0 Residuals are not correlated.

H_1 Residuals are correlated

```
. xtcsd, pesaran abs  
  
Pesaran's test of cross sectional independence =      68.044, Pr = 0.0000  
Average absolute value of the off-diagonal elements =      0.430
```

The p-value from the Pesaran test, 0.0000, is less than a significant level of 5%, so we reject the null hypothesis. As a result, we conclude that serial correlation exists in the model. Because the presence of serial correlation in panel data models biases the standard errors, its results are less efficient.

Conclusion

The gravity equation of bilateral trade flows among 15 European countries was objectively applied in this study to investigate different panel data models. The panel data used by Serlenga and Shin (2007) to investigate the bilateral trade relationship that exists between different trading partners (ninety - one country pairs) over a 42-year period was used in this study. Three major approaches to panel data were used to analyse the bilateral relationships that exist between the various country pairs. The pooled regression was estimated using classical assumptions, but the estimates were

likely biased because unobserved heterogeneity was ignored. Fixed effect and random effect models were developed to account for unobserved heterogeneity. The fixed effect estimator assumed that the variables included were correlated with fixed individual effects, while ignoring time-invariant variables. The random effect estimator, on the other hand, took into account both time-varying and time-invariant variables, assuming that individual effects are random and uncorrelated with all explanatory variables.

The Hausman test was used to determine which model is appropriate for the study. Hausman's test concludes that the fixed-effects model is more appropriate for explaining variations in bilateral trade. Diagnostic tests, such as the Breusch-Pagan Lagrange multiplier test and the Modified Wald test for groupwise heteroscedasticity in the fixed-effect model, were also used to assess the model estimates' validity.

Finally, by taking the first difference, the pooled regression was re-estimated. The assumption was that taking the first difference would remove any unobserved heterogeneity, so time-invariant variables were not considered. The Breusch-Pagan Lagrange multiplier test revealed random effects was appropriate, so the pooled result was insignificant in this study. The heteroscedasticity test revealed that the variances of the fixed effect model are not constant, so the model was re-estimated with robust standard errors to account for heteroscedasticity. When compared to other models, the magnitude of the first difference estimator's coefficients was significantly different. The fixed effect estimator, on the other hand, demonstrated that only real exchange rates between European currencies and the US dollar have a negative impact on bilateral trade volumes. After removing time-invariant variables, all other regressors had a positive effect on trade volumes. Both the pooled and random effect models' time-invariant variables had similar effects on bilateral trade volumes.

To summaries, the best estimator is determined by the assumptions underlying each model as well as how the data responds to the assumptions. Despite the fact that the fixed effect model is best suited to this study, it does not account for time-invariant variables. If a researcher wants to know how these time-invariant variables affect total bilateral trade volumes, he or she should use a model that can estimate the coefficients of these variables.

Appendix

Appendix 1 – Coefficient for Pooled, Fixed and Random effect model

	(1) Pooled	(2) Fixed	(3) Random
SIM	0.849*** (0.0172)	1.228*** (0.0567)	1.189*** (0.0478)
RLF	0.0250*** (0.00841)	0.0313*** (0.00787)	0.0332*** (0.00781)
RERT	0.181*** (0.0209)	-0.117*** (0.0249)	-0.0759*** (0.0230)
RER	0.0907*** (0.00392)	0.0840*** (0.00991)	0.0839*** (0.00911)
CEE	0.233*** (0.0244)	0.317*** (0.0161)	0.325*** (0.0160)
GDP	1.544*** (0.0131)	1.951*** (0.0357)	1.880*** (0.0317)
EMU	0.214*** (0.0508)	0.0274 (0.0294)	0.0558* (0.0290)
LAN	0.249*** (0.0341)		0.410** (0.185)
DIST	-0.683*** (0.0226)		-0.530*** (0.118)
BOR	0.531*** (0.0338)		0.437** (0.191)
Constant	-10.93*** (0.245)	-19.48*** (0.395)	-15.11*** (0.959)
N	3822	3822	3822
Standard errors in parentheses			
* p<0.1, ** p<0.05, *** p<0.01			

Appendix 2 – Stata Output for Pooled regression

```
.
. reg TRADE SIM RLF RERT RER CEE GDP EMU LAN DIST BOR
```

Source	SS	df	MS	Number of obs	=	3,822
Model	12498.0819	10	1249.80819	F(10, 3811)	=	3702.94
Residual	1286.28181	3,811	.337518187	Prob > F	=	0.0000
				R-squared	=	0.9067
				Adj R-squared	=	0.9064
Total	13784.3637	3,821	3.60752779	Root MSE	=	.58096

TRADE	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
SIM	.8494035	.0171976	49.39	0.000	.815686	.883121
RLF	.0249879	.0084132	2.97	0.003	.008493	.0414828
RERT	.1813633	.0209483	8.66	0.000	.1402924	.2224342
RER	.0906828	.0039186	23.14	0.000	.0830001	.0983655
CEE	.233327	.0244463	9.54	0.000	.1853978	.2812562
GDP	1.543887	.0130767	118.06	0.000	1.518249	1.569525
EMU	.2137038	.0507735	4.21	0.000	.114158	.3132496
LAN	.2487985	.0340738	7.30	0.000	.1819939	.3156031
DIST	-.6828109	.0226465	-30.15	0.000	-.7272113	-.6384105
BOR	.5311423	.0337685	15.73	0.000	.4649363	.5973482
_cons	-10.93177	.2446889	-44.68	0.000	-11.4115	-10.45203

```
. eststo Pool
```

Appendix 3 – Stata Output for Random effect model

. xtreg TRADE SIM RLF RERT RER CEE GDP EMU LAN DIST BOR, re						
Random-effects GLS regression			Number of obs		=	3,822
Group variable: id			Number of groups		=	91
R-sq:			Obs per group:			
within = 0.8982			min =			42
between = 0.8974			avg =			42.0
overall = 0.8952			max =			42
			Wald chi2(10)		=	33466.38
corr(u_i, X) = 0 (assumed)			Prob > chi2		=	0.0000
TRADE	Coef.	Std. Err.	z	P> z	[95% Conf. Interval]	
SIM	1.18895	.0477741	24.89	0.000	1.095315	1.282585
RLF	.0332146	.0078101	4.25	0.000	.0179071	.0485221
RERT	-.0758606	.0230384	-3.29	0.001	-.121015	-.0307062
RER	.0839463	.0091093	9.22	0.000	.0660925	.1018001
CEE	.3246798	.0159505	20.36	0.000	.2934174	.3559422
GDP	1.880274	.0317116	59.29	0.000	1.81812	1.942428
EMU	.055788	.0290003	1.92	0.054	-.0010516	.1126276
LAN	.4099014	.1852032	2.21	0.027	.0469099	.772893
DIST	-.5303253	.1184368	-4.48	0.000	-.7624573	-.2981934
BOR	.4366103	.1907719	2.29	0.022	.0627042	.8105164
_cons	-15.11136	.9592534	-15.75	0.000	-16.99146	-13.23126
sigma_u	.51584352					
sigma_e	.29210071					
rho	.75720372	(fraction of variance due to u_i)				

Appendix 4 – Stata Output for Fixed effect model

```
. xtreg TRADE SIM RLF RERT RER CEE GDP EMU,fe
```

Fixed-effects (within) regression

Group variable: id

Number of obs = 3,822

Number of groups = 91

R-sq:

within = 0.8983

between = 0.7987

overall = 0.8204

Obs per group:

min = 42

avg = 42.0

max = 42

F(7,3724) = 4699.29

Prob > F = 0.0000

corr(u_i, Xb) = -0.1343

TRADE	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
SIM	1.227669	.0566658	21.67	0.000	1.11657	1.338768
RLF	.0313344	.0078703	3.98	0.000	.0159039	.046765
RERT	-.1165068	.0249083	-4.68	0.000	-.165342	-.0676715
RER	.0839729	.0099129	8.47	0.000	.0645376	.1034082
CEE	.3165066	.0160553	19.71	0.000	.2850285	.3479848
GDP	1.951331	.0356599	54.72	0.000	1.881416	2.021246
EMU	.02743	.0294323	0.93	0.351	-.030275	.085135
_cons	-19.47683	.3951507	-49.29	0.000	-20.25156	-18.7021
sigma_u	.76239634					
sigma_e	.29210071					
rho	.87199739	(fraction of variance due to u_i)				

F test that all u_i=0: F(90, 3724) = 232.87

Prob > F = 0.0000

```
. eststo Fixed
```

Appendix 5 – Stata Output for first difference

. regress d.TRADE d.SIM d.RLF d.RERT d.RER d.CEE d.GDP d.EMU						
Source	SS	df	MS	Number of obs	=	3,731
				F(7, 3723)	=	212.08
Model	24.553761	7	3.50768015	Prob > F	=	0.0000
Residual	61.5751315	3,723	.016539117	R-squared	=	0.2851
				Adj R-squared	=	0.2837
Total	86.1288925	3,730	.023090856	Root MSE	=	.1286
D.TRADE	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
SIM D1.	.8231778	.1030508	7.99	0.000	.6211363	1.025219
RLF D1.	.0003837	.0049203	0.08	0.938	-.009263	.0100304
RERT D1.	-.1946392	.0397164	-4.90	0.000	-.2725072	-.1167712
RER D1.	.4098286	.0242525	16.90	0.000	.3622791	.4573782
CEE D1.	.0709988	.0147488	4.81	0.000	.0420823	.0999152
GDP D1.	2.004045	.092452	21.68	0.000	1.822784	2.185307
EMU D1.	-.029149	.017592	-1.66	0.098	-.0636399	.0053419
_cons	-.0005899	.0038741	-0.15	0.879	-.0081855	.0070057

Appendix 6 – Stata Output for Hausman Test

```
. hausman Fixed Random //Hausman test
```

	—— Coefficients ——		(b-B) Difference	sqrt(diag(V_b-V_B)) S.E.
	(b) Fixed	(B) Random		
SIM	1.227669	1.18895	.038719	.0304738
RLF	.0313344	.0332146	-.0018801	.0009717
RERT	-.1165068	-.0758606	-.0406462	.0094687
RER	.0839729	.0839463	.0000266	.00391
CEE	.3165066	.3246798	-.0081732	.0018318
GDP	1.951331	1.880274	.0710573	.0163096
EMU	.02743	.055788	-.028358	.005024

b = consistent under Ho and Ha; obtained from xtreg
 B = inconsistent under Ha, efficient under Ho; obtained from xtreg

Test: Ho: difference in coefficients not systematic

$\chi^2(7) = (b-B)'[(V_b-V_B)^{-1}](b-B)$
 = 29.49
 Prob>chi2 = 0.0001
 (V_b-V_B is not positive definite)

Appendix 7 – Heteroscedasticity Test

```
. xttest3 //heteroscedasticity test for FEM  
  
Modified Wald test for groupwise heteroskedasticity  
in fixed effect regression model  
  
H0:  $\sigma(i)^2 = \sigma^2$  for all  $i$   
  
chi2 (91) = 6696.19  
Prob>chi2 = 0.0000
```


Appendix 8 Estimated Fixed Effect Model under Heteroscedasticity.

```
. xtreg TRADE SIM RLF RERT RER CEE GDP EMU LAN DIST BOR,fe
note: LAN omitted because of collinearity
note: DIST omitted because of collinearity
note: BOR omitted because of collinearity

Fixed-effects (within) regression              Number of obs   =       3,822
Group variable: id                           Number of groups =        91

R-sq:                                         Obs per group:
    within = 0.8983                          min =           42
    between = 0.7987                         avg =          42.0
    overall = 0.8204                         max =           42

corr(u_i, Xb) = -0.1343                      F(7,3724)       =      4699.29
                                           Prob > F        =       0.0000
```

TRADE	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
SIM	1.227669	.0566658	21.67	0.000	1.11657	1.338768
RLF	.0313344	.0078703	3.98	0.000	.0159039	.046765
RERT	-.1165068	.0249083	-4.68	0.000	-.165342	-.0676715
RER	.0839729	.0099129	8.47	0.000	.0645376	.1034082
CEE	.3165066	.0160553	19.71	0.000	.2850285	.3479848
GDP	1.951331	.0356599	54.72	0.000	1.881416	2.021246
EMU	.02743	.0294323	0.93	0.351	-.030275	.085135
LAN	0	(omitted)				
DIST	0	(omitted)				
BOR	0	(omitted)				
_cons	-19.47683	.3951507	-49.29	0.000	-20.25156	-18.7021
sigma_u	.76239634					
sigma_e	.29210071					
rho	.87199739	(fraction of variance due to u_i)				

F test that all u_i=0: F(90, 3724) = 232.87 Prob > F = 0.0000

References

- Eichengreen, B. and D. A. Irwin, 1998, “The Role of History in Bilateral Trade Flows,” in J. A. Frankel, Ed., *The Regionalization of the World Economy*, University of Chicago Press
- Hsiao, C. (2007). Panel data analysis—advantages and challenges. *Test*, 16(1), 1-22.
- Hsiao, C. (2014). *Analysis of panel data* (No. 54). Cambridge university press.
- Mátyás, L., 1997, “Proper Econometric Specification of the Gravity Model,” *The World Economy*, 20, 363-368.
- Pillai, V. N. (2017). Panel Data Analysis Lecture Notes, Econometrics. Retrieved from: https://www.researchgate.net/publication/319629513_Panel_Data_Analysis_Lecture_Notes
- Serlega, L. and Y. Shin (2007) “Gravity models of intra-EU trade: Application of the CCEP-HT estimation in heterogeneous panels with unobserved common time-specific factors.” *Journal of Applied Econometrics* 22, 361-81.